

2015 Porcupine River Chum Salmon Telemetry Program



Prepared For
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Down to Earth Biology

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EXECUTIVE SUMMARY

In 2011, the Vuntut Gwitchin Government (VGG) began the enumeration of Porcupine River chum salmon via sonar at a site in the near vicinity of Old Crow. Prior to 2011, chum salmon in the Porcupine River had been enumerated at a counting weir operated by the Department of Fisheries and Oceans on the Fishing Branch River, located approximately 350 km upstream of Old Crow. In 2011 and 2012 concurrent weir and sonar enumerations were completed so that future sonar based escapement estimates could be compared to historical weir based estimates. Radio telemetry studies on chum salmon were completed by VGG in 2013 and 2014 to provide additional comparative data to allow for a better understanding of the relationship between weir and sonar counts. In 2015 VGG undertook a similar radio tracking program to provide a 5th year of comparative data and better understand spawning locations in the Porcupine River watershed.

Esophageal implant radio tags were applied to 95 chum salmon during late August and September, 2015, at the chum sonar site at Old Crow. Telemetry tracking flights were conducted from October 15 – 18 and November 3 – 4 to relocate tagged chum salmon at spawning sites in the Canadian portion of the Porcupine River watershed.

In total, 90 of the 95 tags were accounted for through aerial tracking and/or by data collected by the two stationary telemetry towers. Four tags were recaptured and turned in by local fishers to the VGG Natural Resources Department, after the end of the field program. Eight tags were located in the vicinity of Old Crow or other mainstem areas where spawning is unlikely to occur. One chum was shown to have migrated downstream after radio tag application by a fixed telemetry tower located downstream of the tagging location. These 13 tags, in addition to the 5 that were not located, resulted in 77 tags that could have migrated above the weir site.

Of the 77 radio tagged chum salmon that were relocated on the spawning grounds, the largest proportion (74%) were relocated upstream of the Fishing Branch River weir. The stationary radio receiver located at the weir site did record some tag movement downstream towards the weir 10-14 days after their initial upstream passage; however, none of these tags (presumably post-spawn mortalities) drifted downstream past the weir site.

The total 2015 Porcupine River chum salmon sonar passage estimate at Old Crow was 21,397, of which 556 were later captured in the VGG subsistence fishery upstream of the sonar site. Captured chum salmon were removed from the total sonar count, resulting in an adjusted escapement estimate of 20,841. Multiplying this escapement estimate by the proportion of chum tags relocated above the weir provides an estimate of the total number of chum that spawned in the upper Fishing Branch River; this data suggests that approximately 15,151 chum salmon may have spawned upstream of the weir in 2015, including periods before and after the weir was operational. This estimate is considerably larger than the 2015 weir passage estimate of 8,531 and may be related to several factors (i.e. fish passage before and after the weir was installed, over estimate of chum passage at the sonar site, tagging of proportionally higher amount of Fishing Branch river chum).



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ACKNOWLEDGEMENTS

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Local VGFN citizen Robert Kyikavichik assisted with radio tag applications and provided the practical input to ensure the project's success.

AUTHORSHIP

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1 INTRODUCTION

1.1 PROJECT BACKGROUND AND GOALS

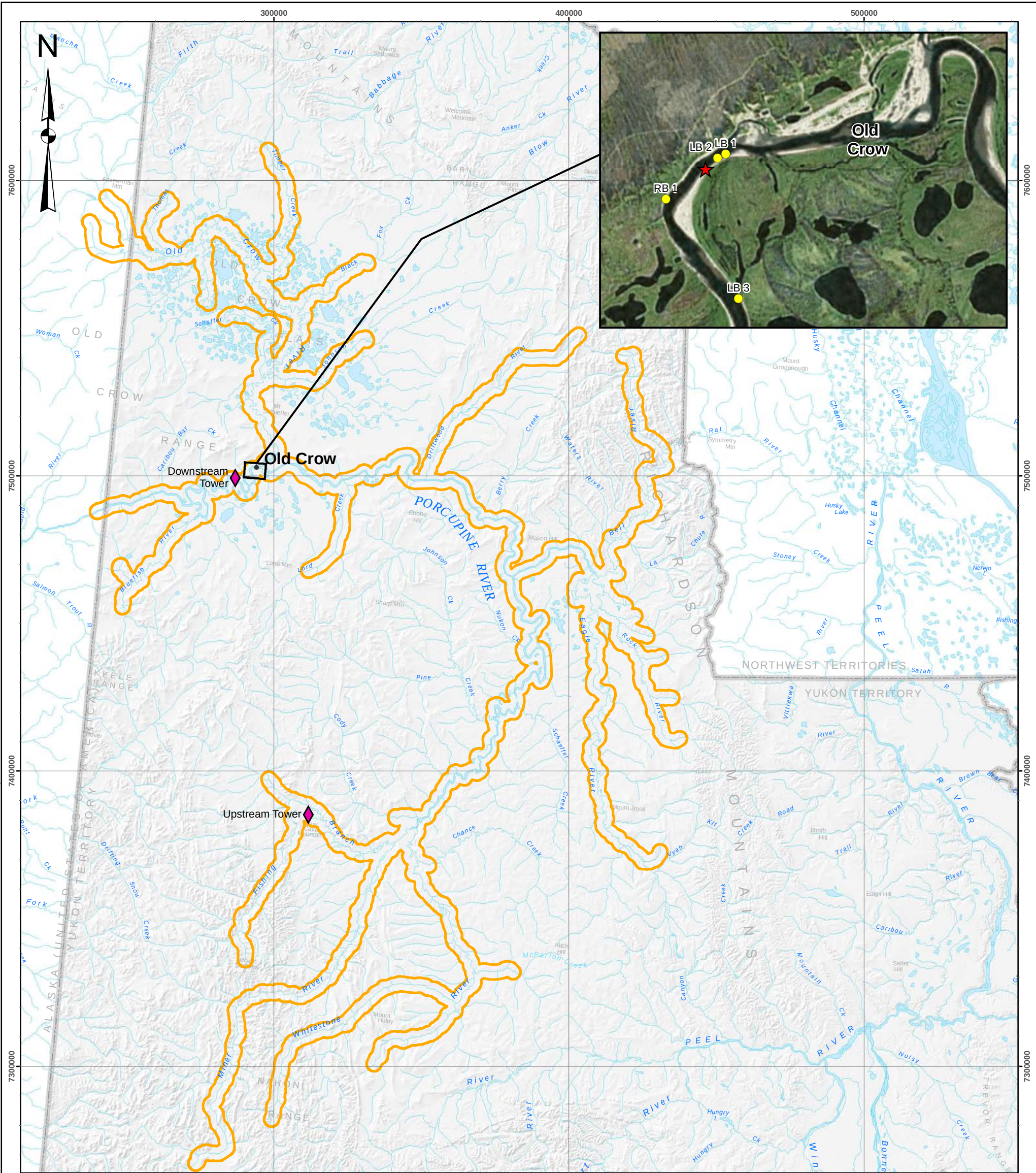
Salmon migrating up the Porcupine River through the Traditional Territory of the Vuntut Gwitchin Government (VGG) are culturally important to VGG citizens. Old Crow, the only Canadian community on the Porcupine River, relies on the salmon fishery as a traditional food source. The subsistence harvest is also an important component in the traditional lifestyle of Old Crow residents.

The Yukon River Panel was established to manage “transboundary” salmon stocks, which move through waters subject to both U.S. and Canadian fisheries management processes, in a cooperative forum. Three salmon species spawn in the Canadian portion of the Porcupine River: Chinook (*Oncorhynchus tshawytscha*), chum (*Oncorhynchus keta*) and coho (*Oncorhynchus kisutch*). Chinook and chum salmon are managed jointly through the Yukon River Panel’s Joint Technical Committee (JTC) process; however, there is currently no joint management process for coho salmon on the Porcupine River.

In 2009, VGG initiated the Porcupine River sonar program, to allow for a more accurate in-season enumeration of chum salmon at a site located 2 km downstream of Old Crow. The Porcupine River sonar program has provided chum salmon escapement estimates annually since 2011. Prior to 2011, chum salmon in the Porcupine River had been enumerated at a counting weir operated by the Department of Fisheries and Oceans (DFO) on the Fishing Branch River, located approximately 350 km upstream of Old Crow.

In 2013, VGG initiated the Porcupine River Chum Salmon Telemetry program to facilitate the transition from weir to sonar based escapement goals; this program was intended to quantify the proportion of chum salmon migrating past the sonar site which spawn upstream of the former weir site. The concurrent operation of the sonar and the Fishing Branch River weir in 2011 and 2012 provided two years of comparative data in this regard. In 2013 and 2014, no weir counts were available but radio tagging of chum salmon at the sonar site, and subsequent tracking of these salmon, provided a means to estimate the proportion of fish that were counted on the sonar and were destined for spawning areas upstream of the weir. Based on tag relocations, it was determined that 74.1% and 46.1% of fish that were enumerated at the sonar were destined for spawning areas above the weir in 2013 and 2014, respectively. These proportions were used to estimate what the 2013 and 2014 Fishing Branch River weir count would have been if the weir had remained in place.

The primary goal of the 2015 telemetry program was to add to the data set of comparative weir/sonar counts by providing an estimate of the number of chum that passed the weir site. The results from 2015 provide a fifth year of comparative tagging and/or concurrent sonar/weir data to help transition escapement estimates from the weir to sonar. A secondary goal of this project was to locate and document chum salmon spawning in other locations within the Porcupine River watershed, including portions of the Fishing Branch River downstream of the weir. This program also collected information on chum salmon



Legend

- ★ Approximate Location of Sonar Site
- Radio Tag Application Site
- ◆ Stationary Radio Telemetry Tower Site
- Extent of 2015 Aerial Telemetry Surveys

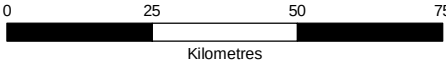
Overview of 2015 Porcupine River tag application sites and extent of telemetry surveys

Data Sources
1,000,000 Topographic Spatial Data courtesy of Her Majesty the Queen in Right of Canada, Natural Resources Canada. All Rights Reserved.

1:1,000,000 National Topographic Database (NTDB) provided by Geomatics Yukon - Yukon Government via online source (Corporate Spatial Warehouse) www.geomatics.yukon.ca.

Project data displayed is site specific. Data collected by EDI Environmental Dynamics Inc. (2015) was obtained using Garmin GPS technology.

Disclaimer
This document is not an official land survey and the spatial data presented is subject to change.



Map Scale 1:1,300,000 (printed on 11 x 17)
Map Projection: North American Datum 1983 CSRS Yukon Albers

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Date:
03/02/2016

MAP 1



Vuntut Gwitchin First Nation





spawning habitat in these areas, and helped to develop local capacity to conduct fisheries work within the community of Old Crow.

1.2 RATIONALE FOR TRANSITION TO SONAR BASED ESCAPEMENT ESTIMATES

The primary motivation for the transition of Porcupine River chum salmon stock assessment from weir based to sonar based escapement goals was related to the effective in-season management of this stock by fisheries managers. The sonar site is located downstream of a large proportion of the local VGG chum salmon fishery and therefore provides a more effective means to provide an in-season chum salmon passage data of Porcupine River chum salmon than the Fishing Branch weir. This sonar also provides a means to quantify the proportion of Porcupine River chum salmon that are not destined for spawning grounds upstream of the Fishing Branch weir, which can be a notable portion of the spawning population in some years.

1.3 STUDY AREA

The Porcupine River is one of the largest tributaries of the Yukon River watershed, much of which is located within the Traditional Territory of the VGG. It has a number of large tributaries within Canada, including the Whistone, Miner, and Fishing Branch rivers (Map 1). The only substantial Canadian settlement within the Porcupine River watershed is the community of Old Crow, located approximately 80 kilometres east of the Canada/U.S. border at the confluence of the Old Crow and Porcupine rivers. Old Crow has a population of approximately 300 people, mainly VGG citizens.

During 2015, all radio tags were applied in the vicinity of the Porcupine River sonar site (Map 1). No tags were applied upstream of Old Crow. Radio tag tracking flights were flown over much of the Canadian portion of the Porcupine River watershed (Map 1) and included all major tributaries including the Miner, Whistone, Fishing Branch (including the North Fork), Bell and Old Crow rivers.

Other medium sized tributaries that were surveyed included the Driftwood, Eagle, Rock and Bluefish rivers. A number of smaller tributaries were also surveyed including David Lord, Black Fox, Timber, Cody, Chance, McParlon, and Pine creeks. As a new component of the 2015 project, radio receivers were installed in stationary radio towers downstream of Old Crow as well as at the Fishing Branch River weir (Map 1) to collect additional data on the migration of tagged chum salmon (see Section 2.2.2 for additional details on this method).



2 METHODS

2.1 CHUM SALMON RADIO TRACKING

The Porcupine River chum salmon sonar program was operated by DFO from August 8 to September 30, 2015. Radio tags were applied by a biologist from EDI and a local Vuntut Gwitchin First Nation technician from August 19-24 (1 tag), September 6-16 (58 tags) and September 21-26 (36 tags).

Chum salmon were captured via set netting and onshore drift netting. Onshore netting was conducted in the downstream vicinity of the Porcupine River sonar site and set netting was conducted at various sites within 1-2 km of the sonar site (Map 1). Many of these sites had been used during previous chum mark-recapture programs (Snow 2010) and were known to be effective fishing sites at various water levels. Onshore drift netting was conducted along the shore of the Porcupine River; one end of the net was walked downstream from shore while the other was secured to a boat which was slowly drifting downstream at the same speed. Set nets were constantly monitored and checked regularly. All nets were 100 feet in length with a net mesh depth equivalent to approximately 10 feet; all nets were hung at a ratio of 3:1. Nets were single panels, with mesh sizes of 6.0 and 6.75 inches (15 and 17 cm, respectively).

Fish captured in drift/set nets were quickly removed from the net and placed in a water filled tote to recover. Fish were observed for any signs of injury (e.g. lacerations, bleeding from gills), and only healthy fish were selected for tagging. Scale samples were collected from all captured chum (five scales per individual) and delivered to DFO for processing; any injured chum salmon were sexed, measured, and released untagged. All other captured fish were identified to species, measured and released. Genetic samples were collected from radio tagged chum salmon by collecting paired axillary processes, which were immersed in ethanol filled vials in the field in order to preserve them. A separate vial was used for each sampled chum salmon, and the tag frequency and code of the implanted radio tag was recorded on the sample vial for each individual. Filled sample vials were delivered to DFO for analysis and processing at the completion of the field program.

Chum salmon selected for tagging were fitted with ATS (Advanced Telemetry Systems) model F1840B esophageal implant tags. In addition to the radio tag, a secondary numbered orange “spaghetti” (or floy) tag was applied to each radio tagged chum (Photo 1). Spaghetti tags were meant to make radio tagged fish more visible to local fishers; tags were brightly colored and fishers were accustomed to looking for tags due to previous chum salmon research programs on the Porcupine River. Following each successful tag placement, fish were allowed to recover once more and were inspected for any signs of injury or unusual stress prior to release. Radio tags were removed from chum that showed signs of injury or abnormal behaviour after tagging and were reapplied to a different individual.



Photo 1. EDI field crew member with a male chum salmon tagged with an esophageal implant tag and posterior secondary floy tag; September 14, 2015.

A total of 95 radio tags on three different frequencies were utilized for this project, as shown in Table 1. One additional tag was retained as a test tag to allow for the testing of the radio receiver and antennas prior to tracking flights. Each tag was coded with a unique identifier; a full list of tag frequency and unique tag identifiers is provided in Appendix A. Tags were also encoded with a mortality sensor, to help determine whether tracked chum had reached their terminal spawning destinations. Mortality sensors were pre-set to activate if no movement was registered by the tag for 24 hours or more.

Table 1. Frequencies of radio tags applied to chum salmon at the Porcupine River sonar site in 2015.

Tag Frequency	Number of Tags
150.243	31
150.283	32
150.323	32



2.2 TRACKING OF RADIO TAGGED CHUM SALMON

2.2.1 AERIAL TRACKING

Aerial tracking of radio tagged chum was conducted from October 15 to 18 and November 3 to 4, 2015. The areas flown on each day of the survey are shown in Table 2. Key spawning areas, such as the Fishing Branch River mainstem, were flown during both tracking events to maximize survey time and ensure that all tags were detected.

Table 2. Summary of chum salmon telemetry surveys in the Porcupine River Watershed in 2015.

Date of Tracking Flight	Survey Locations
October 15, 2015	- Southern half of the Miner River - South and North fork of the Fishing Branch River, including three passes between Bear Cave mountain and the weir location - North fork of the Whitestone River, including McParlon Creek - David Lord Creek - Porcupine River mainstem from David Lord Creek to Canada/U.S. border - Bluefish River
October 16, 2015 ^a	- Porcupine River mainstem from Old Crow to Bell River - Bell River, including the southern portion of Little Bell River - Northern portions of the Rock and Eagle Rivers - Driftwood River - Crow River, including the eastern portion of Johnson, Black Fox, Timber and Thomas creeks
October 18, 2015	- Porcupine River mainstem from Bell River to Miner River - Southern portion of the Miner River
November 3, 2015	- South Western portion of the Whitestone River - Southern half of Eagle River - Rock River - Western portion of Bell River - Mainstem of the Porcupine from the confluence of the Bell to the Bluefish River - Northern portion of the Bluefish River
November 4, 2015	- Crow River, including Schaeffer, Surprise and Bilwaddy creeks - Southern portion of the Porcupine mainstem - Fishing Branch River, North and South fork

^aAerial tracking flights not conducted on October 17, 2015 due to poor weather conditions.

Aerial tracking was conducted by an EDI biologist; all tracking was completed in a Cessna 172 chartered from Whitehorse. Radio antennas were mounted to each wing strut of the plane, to allow for optimal detection of tags on either side of the plane's flight path. Tracking was completed using an ATS model R4520C GPS equipped receiver; the same receiver unit was used on all flights. The frequencies of applied tags were inputted into the receiver, and the receiver was set to cycle through each frequency at a scan rate of 3 seconds per channel.

While tracking, the plane was flown as low as possible (300 to 500 feet), at speeds ranging from 100 to 130 km/h (the slowest possible speed, dependent on wind conditions). When a tag tone was heard by the technician, the frequency was locked using the receiver's "hold" function and the tag was decoded. The



telemetry recorded the GPS position of each tag automatically. When required, the pilot was asked to fly a loop to allow the receiver sufficient time to detect and decode the tag. On October 15th, the Fishing Branch River (where many tagged chum were expected to have spawned) was flown three times, with one frequency on 'hold' for each pass. This facilitated decoding of all tags in the area, and helped ensure that no tags were missed due to the receiver cycling through frequencies.

2.2.2 STATIONARY RADIO TOWER TRACKING

In order to gain a better understanding of fish movement following tagging events, two stationary (fixed) telemetry towers were installed in 2015 prior to the commencement of radio tagging. The first tower was located approximately 20 km downstream of the tagging location and installed in order to determine if radio tagged chum salmon were migrating downstream after tagging. The second stationary tower was installed adjacent to the weir on Fishing Branch River, approximately 350 km upstream of the tagging location. The purpose of this tower was to identify how many fish spawned above the weir and determine if fish were floating downstream past the weir after spawning.

Each stationary tower was equipped with a solar panel and powered with a 12 volt deep cycle battery. In order to determine the direction of travel of a radio tag that passed by the tower, two antennas were installed on each tower, one pointing upstream, the other pointing downstream. Each tower was equipped with an R4520C ATS telemetry receiver set to stationary scan mode which continuously scanned through all three possible tag frequencies and recorded the date, time and signal strength of detected radio tags. This receiver was also equipped with an antenna switch, to allow for differentiation of tag signals from each antenna.

The upstream tower was deployed on August 24, 2015, several weeks ahead of the first tag application. Data from this tower was set up to be uploaded via satellite modem directly to the EDI office by means of an ATS brand Remote Data Platform (RDP). All data was also set to be backed up on the telemetry receiver's internal memory. Installation of the stationary tower adjacent to the weir on Fishing Branch River not only provided count and timing location of fish passage through the weir, it also enabled the determination of travel time (km/day) of fish from the tagging site to the spawning grounds upstream of the weir.

The downstream tower was installed on August 18, 2015 at the start of the radio tagging program. Data from this tower was set to be backed up on the receiver's internal memory and was retrieved on September 27, 2015 at the end of the program. Periodic checks of the tower and receiver were made by field staff on August 23, September 6 and September 21, 2015 to ensure it was functioning properly.

2.3 DETERMINING LOCATIONS AND MORTALITY

The locations of radio tagged chum salmon were determined using recorded telemetry data collected during radio tracking flights as well as from the two stationary telemetry towers. During tracking flights, the R4520C radio receiver recorded the date, time, frequency, tag identifier, signal strength and GPS coordinates



(UTMs) of each tag that was decoded. After each survey, this data was downloaded from the receiver and exported to Microsoft Excel for further analysis. Data from all days of the survey was compiled together for further analysis. Telemetry data collected by the fixed towers was analyzed in a similar manner, to determine passage dates at the Fishing Branch River weir and the potential for downstream migration of chum salmon after tagging.

Each tag was set to continually broadcast its data, and therefore several data records were collected for most tags. To determine the final position of each tagged chum salmon, all radio tag tracking data (including aerial surveys and stationary radio tower detections) was sorted by frequency, tag identifier and signal strength. For each tag, the data record with the highest signal strength was taken as the best approximation of the tagged fish's location. To determine the status of a tag's mortality sensor, an average of recorded mortality codes were taken for each fish. If a tagged chum salmon was detected multiple times, only the data from the final detection was analyzed. The radio tag mortality sensor outputs one of three possible numerical codes, based on the amount of motion of a tagged fish:

0 = tagged fish in constant motion

1 = intermittent motion of tagged fish

2 = no motion of tagged fish

When averages were taken, every individual with an average mort code of less than one was assumed to be "alive", while averages greater than one are assumed "dead". Individuals with a mean mort code of one were categorized as "unclear".

In most cases, the mortality sensors provide a good indication of whether a tagged fish is alive or dead; however, a false positive is possible. For example, a chum carcass that is caught on large woody debris in an area of high current could be moving enough to produce a mortality code of '0', erroneously suggesting this fish was still alive. For the purposes of this project, the status of the mortality sensor on tagged chum salmon were taken as an indicator of the vitality of each tagged chum, but were not taken as proof that a particular chum salmon was alive or dead.



3 RESULTS

3.1 CHUM SALMON TAGGING

3.1.1 FISH CAPTURE DATA

A summary of daily onshore drift and set netting catches conducted over the duration of the sonar program is presented in Table 3 and Appendix B. A total of 146 fish were captured including 120 chum salmon, 3 Chinook, 1 coho, and 22 freshwater fish. All fish, with the exception of one female chum captured on September 11, were caught using set nets. The aforementioned chum was captured during an onshore drift set at tagging site LB3 (Map 1).

Table 3. Summary of 2015 onshore drift and set netting fish captures by tagging period.

Program Week	Number of Net Sets	Set Netting Effort (Hours)	Chum Salmon Captured	Chinook Salmon Captured	Coho Salmon Captured	Other Fish Species Captured	Total Fish Captured
Aug.19-Aug.24	9	54.9	1	1	0	0	2
Sep.6-Sep.16 ^a	16	127.4	64	2	0	9	75
Sep.21-Sep28	12	47.71	55	0	1	13	69
Total	37	230.0	120	3	1	22	146

^aFishing effort from Sep.6-16 includes 1.5 hours effort spent onshore drift netting which resulted in the capture of 1 female chum

3.1.2 RADIO TAG DEPLOYMENT DATA

A summary of chum salmon radio tag deployments during each tagging period is shown in Table 4. Tag application rates were based upon run timing from previous years and daily chum salmon passage rates at the Porcupine River sonar location. Scheduled tagging periods were adjusted in season when necessary. Detailed information on the location and date that each tag was applied, can be found in Appendix B.

Table 4. Summary of chum salmon radio tag deployments on the Porcupine River in 2015 by tagging period.

Tagging Period	Number of Radio Tagged chum	Proportion of Total Tags (%)
August 19 – August 24	1	1
September 6 – September 16	58	61
September 21 – September 28	36	38
Total	95	100.0

The sex ratios and fork lengths of chum salmon captured between August 19 and September 28, 2015 are shown in Table 5. The average fork length of captured male chum salmon was 69.2 cm, while the average fork length of female chum salmon was 64.8 cm (Table 5). Male chum salmon were tagged more frequently than females, and accounted for 81% of all tagged chum. The lengths and sex ratios of chum salmon that were captured in 2015 are similar to the results of past chum salmon netting programs on the Porcupine



River (EDI 2011, Snow 2010). At the time that this report was written, analysis of scale samples taken from captured chum salmon was not yet completed by DFO.

Table 5. Summary of sex and fork length data from chum salmon captured on the Porcupine River in 2015 by tagging period.

Tagging Period	Male			Female		
	Total Tagged	% of Total	Mean Length (cm)	Total Weekly Tagged	% of Weekly Total	Mean Length (cm)
August 19 – August 24	1	100	71.0	0	0	NA
September 6 – September 16	50	86	69.9	8	14	66.6
September 21 – September 28	26	72	68.2	10	28	63.8
Program Mean	-	81	69.2	-	19	64.8

3.1.3 GENETIC SAMPLE COLLECTION

Genetic samples were taken from 96 chum salmon throughout the three tagging periods. All radio tagged fish were sampled as well as one additional chum salmon mortality. All samples were delivered to DFO for further analysis following the completion of the field program, results are not yet available.

3.2 RELOCATION OF RADIO TAGGED CHUM SALMON

3.2.1 AERIAL TRACKING OF RADIO TAGGED CHUM

Unlike previous years where only one round of telemetry tracking flights were completed, two rounds of chum salmon radio tag tracking flights were conducted in 2015. The second round of flights provided the opportunity to locate any radio tagged chum salmon that were not picked up during the first round of flights, as well to confirm the final spawning destination of tags that were not located during the first set of flights. In total, 90 of the 95 applied tags were relocated however, only 87 of those tags were found during aerial tracking flight. The remaining three tags were relocated by stationary radio towers. The locations of relocated tags are summarized in Table 6; specific coordinates for each tag can be found in Appendix C. Aerial surveys of the Fishing Branch River located 55 tagged chum salmon upstream of the former weir site (Table 6; Map 2). Seven tags were located in the Fishing Branch River downstream of the weir site and no tags were located in the North Fork of the Fishing Branch River.

In addition to tag relocations in the Fishing Branch River, seven tags were located in the upper Porcupine River mainstem, with the remaining 6 tags detected in the Bell River, Bilwaddy Creek and Miner River. An additional four tags were captured by local fisherman after the last tagging period.



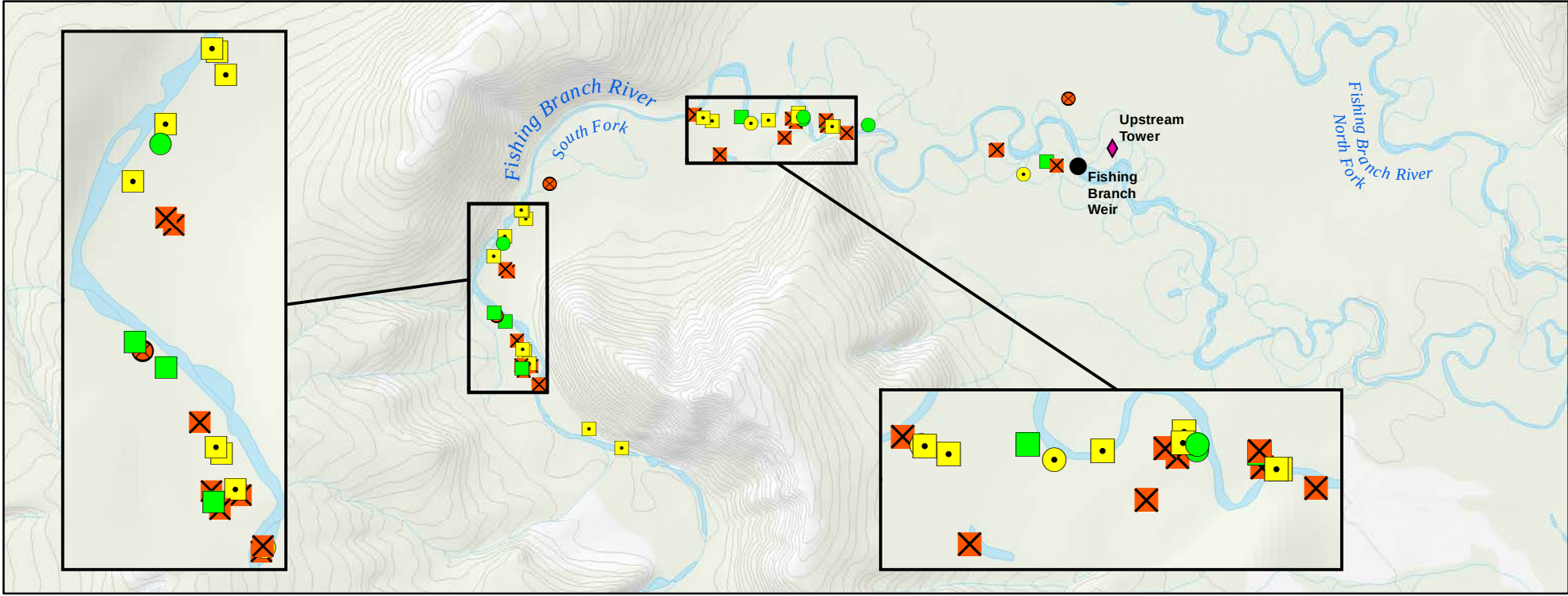
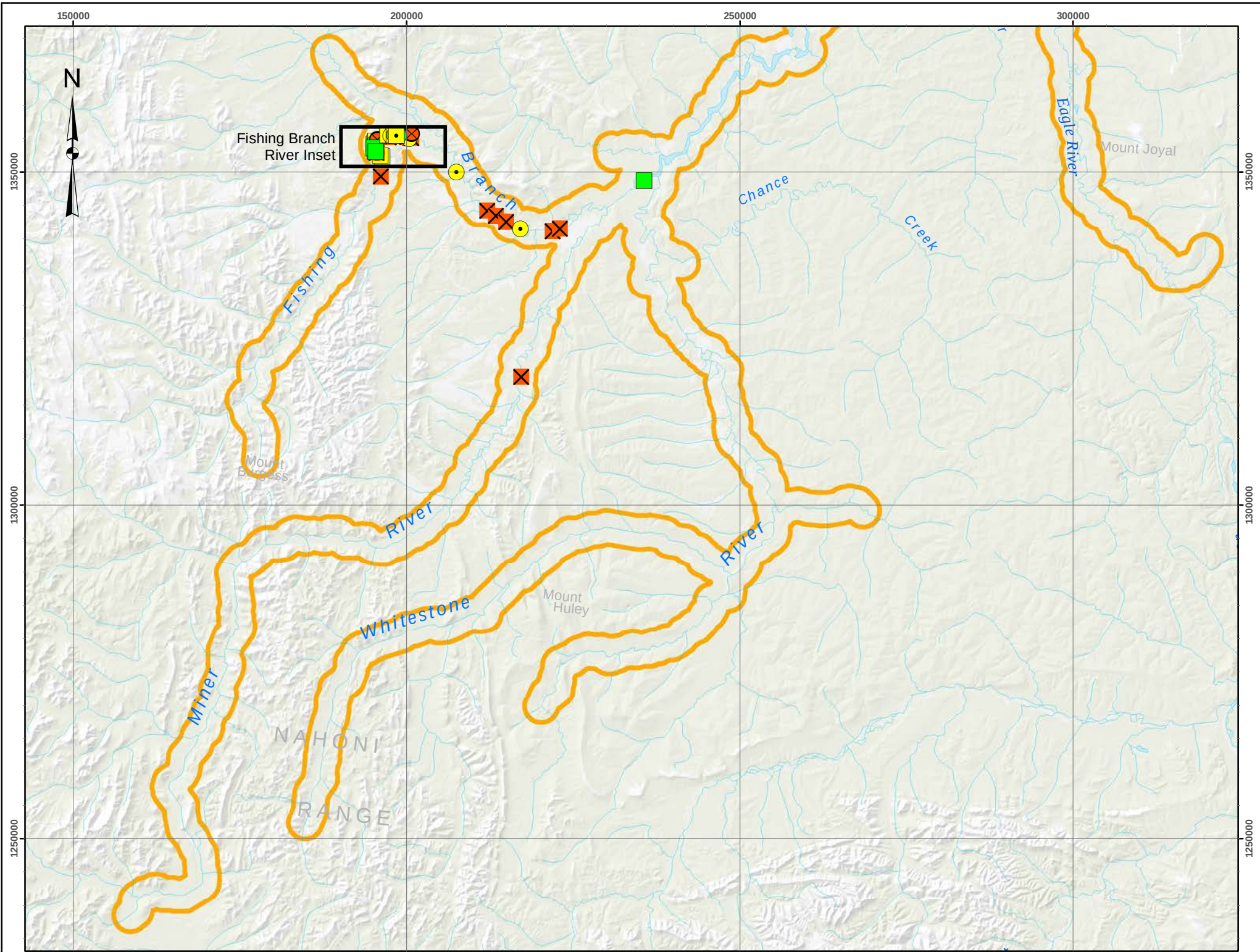
3.2.2 TRACKING OF RADIO TAGGED CHUM BY STATIONARY RADIO TOWER

The addition of two fixed telemetry towers to the chum salmon telemetry program in 2015 provided valuable insight and data into the location and movement of radio tagged chum throughout the Porcupine River watershed. The tower at the Fishing Branch River weir detected all tags that were found during aerial tracking flights as well as two additional radio tags that were not previously located (during aerial surveys). The tower detected the upstream passage of these two tags on September 12th and 15th, respectively. No tagged chum salmon were detected moving downstream past the tower at the Fishing Branch River weir. The stationary radio tower downstream of the tagging site picked up one tag that was not located during aerial surveys.

Table 6. Summary of locations of radio tagged chum salmon through aerial and fixed tower telemetry data in the Porcupine River watershed in 2015

Radio Tag Location	Number of Tags
South Fork of the Fishing Branch River, upstream of the former weir site	57
South Fork of the Fishing Branch River, downstream of the former weir site	7
Upper Porcupine River mainstem	7
Old Crow and near vicinity	8
Upper Crow River (Bilwaddy Creek)	1
Miner River	2
Bell River	3
Downstream migrant	1
Captured by local fishers	4
Not located	5
TOTAL	95

Both towers experienced minor technical issues throughout the 2015 chum salmon telemetry program. The downstream tower was not operational from September 10 – 21, 2015 due to a power failure. This period partially coincided with the second tagging period (September 6 – 16, 2015). Of the five tags that were not relocated at the end of this project, only one of those tags (150.283 A2; Appendix B) could have been missed, as it was the only missing tag that was applied during the period while the tower was non-operational. The upstream tower experienced a manufacturing defect with the Remote Data Platform (RDP). Shortly after installation, water leaked into the RDP which caused it to fail. No data was lost due to this issue; however, the malfunction prevented the uploading of data via satellite modem which meant data was not available in real-time. The RDP was returned for warranty repairs when the tower was retrieved at the end of the field program.



Legend

- Fishing Branch River Weir
- Stationary Radio Telemetry Tower Site
- Extent of 2015 Aerial Telemetry Surveys
- Location of Radio Tagged Chum Salmon (Sex/Status)**
 - Female - Live
 - Female - Dead
 - Female - Status Unclear
 - Male - Live
 - Male - Dead
 - Male - Status Unclear

Locations of radio tagged chum salmon in the southern portion of the Porcupine River survey area (2015)

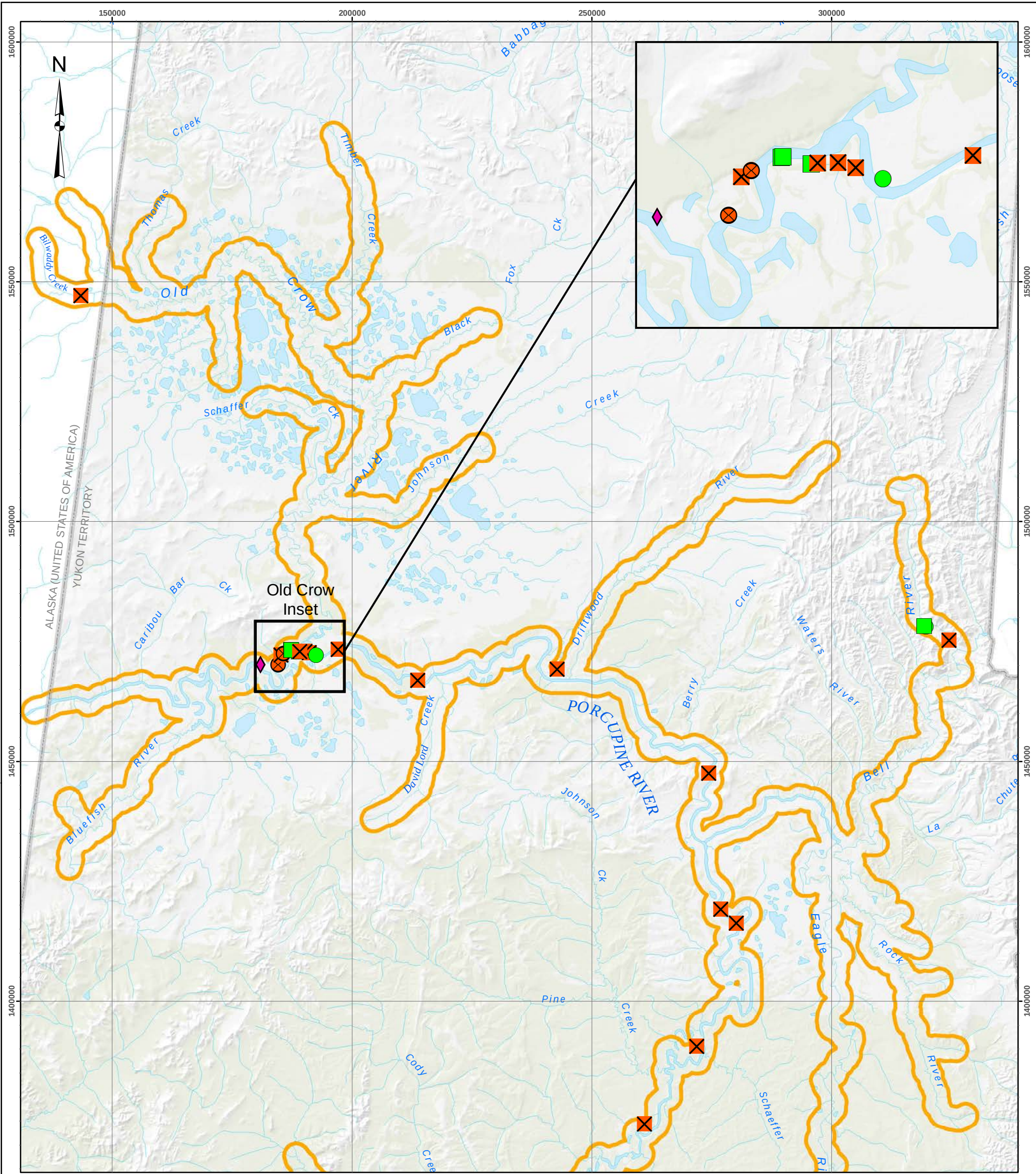
Data Sources
National Topographic Database (NTDB) provided by Geomatics Yukon - Yukon Government via online source (Corporate Spatial Warehouse) www.geomatics.yukon.ca.
Project data displayed is site specific. Data collected by EDI Environmental Dynamics Inc. (2015) was obtained using Garmin GPS technology.

0 1 2 3 4
Kilometres

Reference Scale: 1:650,000 (printed on 11"x17")
Coordinate System: NAD 1983 CSRS Yukon Albers

Drawn: MS	Checked: BSn/BSc	Date: 03/02/2016	MAP 2
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Legend

-  Stationary Radio Telemetry Tower Site
-  Extent of 2015 Aerial Telemetry Surveys
- Location of Radio Tagged Chum Salmon (Sex/Status)**
-  Female - Live
-  Female - Dead
-  Female - Status Unclear
-  Male - Live
-  Male - Dead
-  Male - Status Unclear

Locations of radio tagged chum salmon in the eastern portion of the Porcupine River survey area (2015)

Data Sources

National Topographic Database (NTDB) provided by Geomatics Yukon - Yukon Government via online source (Corporate Spatial Warehouse) www.geomaticsyukon.ca.

Project data displayed is site specific. Data collected by EDI Environmental Dynamics Inc. (2015) was obtained using Garmin GPS technology.



Reference Scale: 1:750,000 (printed on 11"x17")
Coordinate System: NAD 1983 CSRS Yukon Albers

Drawn:
MS

Checked:
BSn/BSc

Date:
03/02/2016

MAP 3





3.2.3 MORTALITY SENSOR STATUS

The status of the mortality sensors of the detected radio tags (excluding tags in the vicinity of Old Crow) is summarized in Table 7. More than half of the detected tags were found with their mortality sensors 'on' (Table 7), meaning that no movement had occurred in over 24 hours. In the lower part of the Fishing Branch River the mortality sensor status of most tags indicated that fish were no longer moving, and these fish were presumed to be deceased. In contrast, the mortality sensors in the tags found in the Porcupine River mainstem and the Bell and Crow rivers showed that most tags were still in motion, and these fish were presumed to be alive or drifting downstream. Given the period of time between tagging and the telemetry flights (a minimum of 19 days), these fish should have been located on the spawning grounds.

Table 7. Summary of mortality sensor status for radio tagged chum salmon in the Porcupine River in 2015 (excludes tagged chum salmon recaptured in local subsistence fishery).

Location	Number of Tags with Mortality Status 'Dead'	Number of Tags with Mortality Status 'Alive'	Number of Tags with Mortality Status 'Unclear'	Percentage of 'Dead' Tags (%)
Fishing Branch River – Upstream of weir site	24	10	19	45%
Fishing Branch River – Downstream of weir site	5	0	2	71%
Porcupine River Mainstem	7	0	0	100%
Bilwaddy Creek	1	0	0	100%
Bell River	1	2	0	33%
Miner River	1	1	0	50%
TOTAL	39	13	21	-

3.3 SPAWNING HABITAT CHARACTERISTICS OF SURVEYED WATERSHEDS

The areas of the Fishing Branch River that are upstream of the weir location are known to be productive chum salmon spawning habitats, and are well documented. The suitability of spawning habitat for chum salmon in the other surveyed tributaries of the Porcupine River was described extensively in the previous 2013 and 2014 Porcupine River Chum Salmon Telemetry reports (EDI 2014b; EDI 2015). Refer to EDI 2014b (Section 4) and EDI 2015 (Section 3.3) for detailed habitat description of the following tributaries: Fishing Branch River North Fork, Miner River, Whitestone River, Porcupine River Mainstem, Bell River, Driftwood River, David Lord Creek, Bluefish River and Crow River. These habitats are not discussed in this report.



4 DISCUSSION

4.1 DEVELOPMENT OF LOCAL CAPACITY

An important goal of the 2015 chum radio tagging program was to continue to develop local capacity to conduct fisheries work within the community of Old Crow. While a number of past programs (e.g., mark-recapture, CPUE index and sonar programs) have helped to develop fisheries technical capacity within the community, this particular program provides several important advantages:

- The program offered approximately 4 weeks of fisheries related work to one individual;
- The skills needed to apply the radio tags and conduct onshore drift and set netting provided new learning opportunities for local technicians; and,
- The program provided a means to employ skilled fisheries technicians in the community, while also providing experience to younger community members who have not been involved in past programs.

One local field technician was trained and participated in the program during 2015. This field technician conducted a portion of the tagging work in conjunction with EDI biologists, and was essential the success of this project. The knowledge of the theory and rationale of radio tagging fish is a specialized skill and there is a considerable amount of interest within the community of Old Crow to pursue further stock assessment programs for chum salmon on the Porcupine River. If the additional radio tagging studies are conducted in the Porcupine River watershed, there is an existing technical understanding of radio tag application techniques that can be leverage to assist with such projects.

4.2 EFFECTIVENESS OF RADIO TRACKING TOWERS

The addition of two stationary telemetry towers provided valuable information to the 2015 chum telemetry program. Both towers were able to show tag movements that aerial tracking flights were not able to detect. The downstream tower showed that one tag migrated downstream after tagging and was therefore removed from the number of tags which could have potentially migrated above the weir site on Fishing Branch River. The tower at the Fishing Branch River weir also detected two radio tags that passed through the weir but were not found during aerial tracking events. Without the tower, these two tags would have gone unaccounted for. The tower at the weir was also able to prove that there was no movement of tagged chum downstream past the weir. If used in the future, the stationary towers would benefit from having a third antenna incorporated. This antenna could be positioned perpendicular to the stream channel and could provide a much more accurate time of passage at the tower. The antenna orientation used during 2015 involved each antenna positioned at a 45° angle to the stream channel and in some cases, this resulted in difficulties in making a determination of the time that each tag passed the tower.



4.3 ADJUSTED TOTAL RADIO TAG PROPORTIONS

The results of the chum salmon radio tag tracking confirm that the Fishing Branch River was the major spawning destination for radio tagged chum salmon in 2015. Of the 95 tags that were applied during the 2015 field program, 57 tags were found in the Fishing Branch River in spawning areas upstream of the DFO enumeration weir. Seven tagged chum salmon were found in spawning areas in the Fishing Branch River downstream of the weir.

A total of four tags were removed by the local VGFN fishery and an additional five tags were not located during the tracking flights. The destinations of these five tags cannot be determined, and they should be removed from the data set when considering the proportion of chum salmon that spawned upstream of the Fishing Branch River weir site. An additional eight tags were located within the vicinity of Old Crow or other mainstem areas, where spawning is unlikely to occur. One tag migrated downstream from the spawning area and was detected by the tower downstream of the tagging site, but not relocated during aerial surveys. Based upon the above details, the remaining 77 chum salmon are considered to have relocated to upstream spawning locations. The adjusted proportion of tagged chum salmon that relocated upstream of the Fishing Branch River weir therefore is 74% (57 of 77 tagged chum; Table 8).

Table 8. Summary of locations of chum salmon radio tagged in the Porcupine River in 2015; tags that were removed in the local fishery, found in the vicinity of Old Crow, migrated downstream or not found during tracking are not included.

Radio Tag Location	Number of Tags Detected	Proportion of adjusted total (%)
South Fork of the Fishing Branch River, upstream of the former weir site	57	74.0
South Fork of the Fishing Branch River, downstream of the former weir site	7	9.1
Upper Porcupine River mainstem	7	9.1
Upper Crow River (Bilwaddy Creek)	1	1.3
Upper Bell River	3	3.9
Miner River	2	2.6
TOTAL	77	-

4.4 ESTIMATE OF CHUM SALMON PASSAGE AT THE FISHING BRANCH RIVER WEIR

With the knowledge of the total number of chum salmon that are estimated to have migrated by the tagging site (via the concurrent DFO chum sonar program), the percentages of radio tags found in a particular area can be used to roughly estimate the numbers of chum salmon at spawning destinations. This apportionment provides a means to estimate the number of fish that spawned above the site of the Fishing Branch River weir in 2015, which was a primary goal of this project.

A total of 21,397 chum salmon were estimated to have passed the Old Crow sonar site in 2015 (MacDonald pers. comm. 2015). However, 556 chum salmon were captured in the local VGG subsistence fishery



upstream of Old Crow (MacDonald pers. comm. 2015). Although some of these chum salmon were undoubtedly not destined for spawning areas upstream of the former weir site, there is no reliable way to partition the local harvest by spawning destination. Removing the entire local harvest upstream of the sonar site from the final sonar count is the best means to adjust the sonar count for harvest. If these chum salmon are removed from the sonar estimate, this results in a chum salmon spawning escapement estimate of 20,841. An estimate of the proportion of this escapement that spawned upstream of the weir is obtained by multiplying 20,841 chum salmon by the adjusted percentage of radio tagged fish that were found upstream of the Fishing Branch weir (74%). This calculation, based solely on radio tag relocations, suggests that an estimated 15,151 chum salmon spawned upstream of the site of the Fishing Branch River weir during 2015.

In 2015, DFO reported a final weir passage estimate for chum salmon of 8,351. The discrepancy between DFO and EDI's estimates of approximately 6,800 chum is considerable; however, there are several factors that could explain the variation. Tagging in 2015 could have unintentionally disproportionately targeted chum destined to spawn upstream of the weir on Fishing Branch River which would cause a higher percentage of spawners found above the weir than in actuality. Another factor that could help explain the discrepancy is the possibility that the majority of tags removed from the total applied number to calculate the Fishing Branch estimate (18 out of 95) were not destined for Fishing Branch. In this case, the estimated proportion of tags that would have spawned above the weir would be closer to 58%. Other factors worth considering are the possibility of the passage of more fish prior to and/or after the weir on Fishing Branch River was operational than calculated by the post-season expanded count. It appears; however, that the most likely explanation for the discrepancy between the estimated report by EDI and the final 2015 weir passage estimate would be a combination of the factors above.

4.5 COMPARISON TO HISTORICAL DATA

In addition to the Fishing Branch River weir, chum salmon stock assessment has been conducted downstream of the community of Old Crow since 2003. This has provided an opportunity to estimate the proportion of the total Canadian origin Porcupine River chum salmon stock that spawns above the weir. From 2003 to 2010, mark-recapture and catch-per-unit-effort (CPUE) index programs operated downstream of Old Crow. These programs provide a means to compare chum salmon passage estimates downstream of Old Crow to estimates generated at the Fishing Branch weir. More recently (2011 to present) chum salmon sonar and telemetry data have been used to compare chum salmon passage at Old Crow to that of the Fishing Branch River weir. Both methods are discussed in more detail in the following sections.

4.5.1 TAG RECOVERY FROM MARK-RECAPTURE/CPUE DATA

The proportion of radio tagged chum salmon that were relocated upstream of the Fishing Branch River in 2015 (74%) can be compared to the historical floy tag recovery rates from the chum salmon Mark Recapture/CPUE index programs (EDI 2011). During the Mark Recapture/CPUE index programs, chum salmon tagging was conducted at a site 23 km downstream of Old Crow; there are no known chum salmon



spawning areas between this site and Porcupine River Sonar site. As shown in Table 9, the tag recovery rate over 7 years of Mark-Recapture and CPUE index programs ranged from 35% to 88%, with an average of 56.4 % (EDI 2011). Although these programs used very different methods from the recent telemetry based studies, the 74% tag detection above the Fishing Branch River weir from the 2015 radio telemetry program is within the range of the Mark-Recapture and CPUE index programs.

Table 9. Relative proportion of floy tagged chum salmon counted at the Fishing Branch River weir (Data Source: EDI 2011).

Year	Tags Counted Through the Fishing Branch River Weir (%) ^A
2003	88
2004	59
2005	71
2006	- ^B
2007	56
2008	45
2009	35
2010	41
Average Tag Recovery Rate	56.4

^A Percentages are calculated after the removal of tags by the Old Crow aboriginal fishery and are corrected for fish that may have crossed the weir while it was not operating.

^B During 2006, the weir was not operated continuously due to flooding and as such a reliable proportion of tags could not be obtained.

4.5.2 2011 TO 2015 WEIR, SONAR AND TELEMETRY DATA COMPARISON

The estimated proportion of chum salmon that were counted at the Porcupine River sonar site and spawned upstream of the Fishing Branch weir in 2015 can be compared to results of the 2014 and 2013 telemetry programs in addition to the concurrent sonar and weir count from 2011 and 2012. Similar to 2011 and 2012, the 2015 Porcupine River sonar operated concurrently with the Fishing Branch River weir. Chum salmon passage estimates from the sonar and weir sites in 2015 are shown in Table 10. The average travel time of salmon from the sonar to the weir site in 2015 was calculated from radio tag detections by the stationary tower at the Fishing Branch weir and was determined to be 14 days; this is comparable to the average travel times of 9 and 13 days found in 2011 and 2012, respectively (EDI 2012, EDI 2013). The 2015 travel time offset was used to compare counts from the sonar and weir programs.

In 2013 and 2014 the Fishing Branch River weir was not operational and chum salmon radio tag relocations upstream of the weir were used in place of weir counts. Despite the difference in the methods of estimation, a similar percentage of the Porcupine River chum salmon sonar passage was estimated to have spawned upstream of the weir site in 2011, 2012, 2013 and 2015 (Table 10). In 2014, there was a notable decrease in the proportion of tagged fish relocated above the former weir site. The 2014 results are consistent with ground observations at the Fishing Branch River spawning site in that year, which indicated that there were very low numbers of spawners during October 2014 (Timpany Pers. Comm. 2015).



Table 10. Comparison of harvest adjusted Porcupine River sonar and Fishing Branch River weir chum salmon passage estimates from 2011 to 2015.

Program Year	Harvest Adjusted Porcupine River Sonar Estimate	Equivalent Fishing Branch River Weir Estimate	Percentage of Sonar Estimate
2011	10,938	8,574	78.3
2012	27,793	21,561	77.5
2013	34,235	25,368	74.1
2014	15,086	7,196	46.1
2015	20,841	15,151	72.7
Averages	21,779	15,570	69.74

4.6 RECAPTURE OF TAGS IN LOCAL FISHERY

The 2015 program had the lowest recapture rates of chum salmon since radio tagging began in 2013; only 4 individuals (4.2% of tagged fish) were captured in the local VGFN subsistence fishery near Old Crow (Table 11). The 2014 radio tagging program saw a relatively similar number of tags recaptured in the local VGFN subsistence fishery (6 tags, or 9.7% of all tagged fish). Comparable recapture rates are perhaps due to the continued use of internal esophageal implant tags and/or differences in fishing locations used by local fishers. Lower capture rates in 2015 may also be directly related to an apparent large decrease in chum harvest by local fishers in 2015, which was due in part to low forecasted chum returns to the Porcupine River in 2015.

Table 11. Comparison of the recapture rate of radio tagged chum salmon to overall capture rates of chum salmon the VGFN subsistence fishery upstream of the Porcupine River sonar site in 2013, 2014 and 2015.

Program Year		Total Count	Number Recaptured in VGFN subsistence fishery	Proportion of Total Count (%)
2013	Chum Salmon Sonar Interpolated Total	35,615	1,380	3.9
	Radio Tagged Chum Salmon	94	29	31
2014	Chum Salmon Sonar Interpolated Total	16,892	1,806	10.7
	Radio Tagged Chum Salmon	95	6	6
2015	Chum Salmon Sonar Interpolated Total	21,397	556	2.6
	Radio Tagged Chum Salmon	95	4	4.2

As was the case in 2013 and 2014, the tagging location of chum salmon in 2015 was in the vicinity of the Porcupine River Sonar site. A small number of local VGFN subsistence fishing sites were also located within this area (approximately 2 km downstream of Old Crow). This tagging site was chosen for several important reasons. First, the primary goal of this project was to estimate the number of chum salmon that spawned upstream of the Fishing Branch River weir site in 2013 and 2014. This estimate was derived from chum sonar counts and tag relocation rates upstream of the weir location. To produce this estimate, the tagging had to be conducted at or near the sonar site. Second, tagging could not be conducted upstream of



the Old Crow River, as it is known to support chum salmon spawning, based on radio tagging detections in 2014 (EDI 2015). If tagging were conducted upstream of the mouth of the Crow River, there would be no way to tag fish that were potentially destined for spawning areas in the Crow River and its tributaries.

4.7 FATES OF MISSING RADIO TAGS

Five radio tagged chum salmon were not detected during aerial surveys or by stationary telemetry receivers during the 2015 program. There are several possible theories that may explain this occurrence. Tag failure could account for a few of the missing tags. Hardware failures on VHF radio tags of the type used in the 2015 program are rare; according to manufacturer data, the failure rate is approximately 2% (Adsem pers. comm. 2013). Although it is understood to be a very infrequent occurrence, a 2% tag failure rate could account for 1-2 of the missing tags that were applied in 2015.

Another possibility, albeit very unlikely, is that a small number of tagged chum salmon may have migrated upstream into un-surveyed areas of the Porcupine River watershed. The telemetry equipment that was used often detected tags a considerable distance (5-10 km) away from the actual tag location. Areas with a high density of tags (Fishing Branch River) were also flown a number of times (once on each frequency) to ensure that all tags were detected. In addition, the incorporation of stationary telemetry towers substantially reduced the uncertainty associated with missing tags that spawned upstream of the Fishing Branch River weir. The installation of a fixed telemetry tower located approximately 20 km downstream of the tagging location refutes the conclusion that a large number of tagged fish migrate downstream after the application of radio tags. Telemetry data showed that only one tagged chum migrated downstream after tagging. Without the introduction of the stationary radio telemetry tower to the 2015 program, the fate of this tag would have been unknown.

Damage to tags could have accounted for some of the missing, for example predators consuming tagged salmon on the spawning grounds and thereby destroying the tags. While this possibility is considered very unlikely, it cannot be fully discounted. The final and most likely explanation for the small number of tags that could not be relocated in 2015 is that they were destined for unknown/unsurveyed portion of the watershed. The Porcupine River watershed drains a vast portion of the northern Yukon, and despite the extensive aerial survey conducted in 2015 it is conceivable the tagged chum salmon migrated to remote areas of the watershed that were not surveyed.



5 CONCLUSION

The 2015 Porcupine River chum salmon telemetry program provided valuable insight into the relationship between sonar counts and the number of chum salmon that spawn upstream of the Fishing Branch River weir location. As shown in 2013 and 2014, telemetry data from 2015 indicates that the primary spawning destination of chum salmon in the Porcupine River watershed is the Fishing Branch River, upstream of the weir location.

In addition, the project provided new information on chum spawning locations in tributaries of the Porcupine River (Bilwaddy Creek and Miner River). To date, the Porcupine River chum salmon telemetry program is one of the most comprehensive projects of its kind ever completed in the Yukon. Not only does the program provide data on inter-annual variations in spawning site selection, it also continues to provide new and useful data on the extent of chum spawning locations in the Porcupine River watershed.

The incorporation of two stationary telemetry towers during the 2015 chum telemetry program provided information not only on the fate of radio tags, it also answered several long debated questions. The downstream tower proved that only a very small portion of the chum tagged throughout the program migrated downstream after being tagged (less than 2%). The installation of the second upstream tower at the Fishing Branch weir site provided valuable information on passage rates and travel time, as well as proving that no fish migrated downstream of the weir site after being detected migrating to spawning sites upstream of the weir.



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Snow. 2010. 2009 Porcupine River Fall Chum Salmon CPUE Index: Yukon Restoration and Enhancement Project CRE-27-09. Prepared by EDI Environmental Dynamics for the Vuntut Gwitchin Government and the Yukon River Panel.

6.1 PERSONAL COMMUNICATIONS

MacDonald, Elizabeth. DFO Aquatic Science Biologist. Personal communication. January 4, 2016.

Timpany, P. Operator of Bear Cave Mountain Eco-Adventures. Discussions with EDI, January 2015.

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**APPENDIX A. TAG FREQUENCY LIST
PROVIDED BY ADVANCED
TELEMETRY SYSTEMS**

Transmitter Data Sheet

Job #: 78651

Ship Date: 7/30/2013

Customer Info

Environmental Dynamics Inc.

Attn: Ben Schonewille

2195 - 2nd Ave

Whitehorse

YT

Y1A 1X8

CANADA

470 FIRST AVE N • ISANTI, MN 55040-7123
763-444-9267 • 763-444-9384 fax
email:sales@atstrack.com • www.atstrack.com

Serial #	Quantity	Model#	Description	Species
01	95	F2050	Ext Mount Transmitter	
		ProgDesc:	Mortality: 24 hr	
Pulse Rate:	45.8 ppm	Pulse Width:	34 ms	Weight: 14.0 gr
				Warranty Life: 65 days
				Battery Cap Life: 131 days
148.250		148.250	148.250	148.250
A-0/830/45		A-1/850/55	A-11/1060/155	A-12/1000/125
148.250		148.250	148.250	148.250
A-13/1090/175		A-14/1130/185	A-15/1140/195	A-16/1160/205
148.250		148.250	148.250	148.250
A-17/1180/215		A-18/1190/225	A-19/1220/235	A-2/890/65
148.250		148.250	148.250	148.250
A-20/1240/245		A-21/1270/255	A-22/1290/265	A-23/1300/275
148.250		148.250	148.250	148.250
A-24/1310/285		A-26/990/115	A-3/910/75	A-4/920/85
148.250		148.250	148.250	148.250
A-5/940/95		A-6/970/105	A-75/1030/145	A-8/1070/165
148.250		148.383	148.383	148.383
A-9/1010/135		A-0/830/45	A-1/850/55	A-11/1060/155
148.383		148.383	148.383	148.383
A-12/1000/125		A-13/1090/175	A-14/1130/185	A-15/1140/195
148.383		148.383	148.383	148.383
A-16/1160/205		A-17/1180/215	A-18/1190/225	A-19/1220/235
148.383		148.383	148.383	148.383
A-2/890/65		A-20/1240/245	A-21/1270/255	A-22/1290/265
148.383		148.383	148.383	148.383
A-23/1300/275		A-24/1310/285	A-26/990/115	A-3/910/75
148.383		148.383	148.383	148.383
A-4/920/85		A-5/940/95	A-6/970/105	A-75/1030/145
148.383		148.383	148.554	148.554
A-8/1070/165		A-9/1010/135	A-0/830/45	A-1/850/55
148.554		148.554	148.554	148.554
A-11/1060/155		A-12/1000/125	A-13/1090/175	A-14/1130/185
148.554		148.554	148.554	148.554

A-15/1140/195	A-16/1160/205	A-17/1180/215	A-18/1190/225
148.554	148.554	148.554	148.554
A-19/1220/235	A-2/890/65	A-20/1240/245	A-21/1270/255
148.554	148.554	148.554	148.554
A-22/1290/265	A-23/1300/275	A-24/1310/285	A-26/990/115
148.554	148.554	148.554	148.554
A-3/910/75	A-4/920/85	A-5/940/95	A-6/970/105
148.554	148.554	148.554	148.859
A-75/1030/145	A-8/1070/165	A-9/1010/135	A-0/830/45
148.859	148.859	148.859	148.859
A-1/850/55	A-11/1060/155	A-12/1000/125	A-13/1090/175
148.859	148.859	148.859	148.859
A-14/1130/185	A-15/1140/195	A-16/1160/205	A-17/1180/215
148.859	148.859	148.859	148.859
A-18/1190/225	A-19/1220/235	A-2/890/65	A-20/1240/245
148.859	148.859	148.859	148.859
A-21/1270/255	A-3/910/75	A-4/920/85	A-5/940/95
148.859	148.859	148.859	
A-6/970/105	A-8/1070/165	A-9/1010/135	

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APPENDIX B. DRIFT AND SET NETTING EFFORT & FISH CAPTURE DATA



2015 Porcupine River Chum Salmon Telemetry Program - Appendix B – Chum Salmon Radio Tagging Data

Table B1. Summary of chum salmon radio tagged during 2015.

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Floy Tag #	Sex	Scale Book Number	Scale Row #	Fork Length (cm)
1	150.243	A-1	23-Aug-15	LB1	4486	male	95471	10-30	71
2	150.243	A-2	8-Sep-15	LB1	4490	male	95640	10-30	67
3	150.243	A-3	8-Sep-15	LB1	4523	male	95640	9-49	68
4	150.243	A-5	8-Sep-15	LB1	3327	male	95640	6-46	71
5	150.243	A-6	8-Sep-15	LB1	4492	male	95640	5-25	78
6	150.243	A-8	8-Sep-15	LB1	4290	male	95640	4-44	72
7	150.243	A-4	8-Sep-15	RB1	3617	male	95640	8-48	73
8	150.243	A-11	9-Sep-15	LB1	3542	male	95640	2-42	71.5
9	150.243	A-13	9-Sep-15	LB1	3723	male	95635	10-50	65
10	150.243	A-14	9-Sep-15	LB1	3149	male	95635	9-49	67
11	150.243	A-15	9-Sep-15	LB1	3260	male	95635	8-48	69
12	150.243	A-16	9-Sep-15	LB1	3261	male	95635	7-47	72
13	150.243	A-17	9-Sep-15	LB1	3724	male	95629	10-50	67
14	150.243	A-9	9-Sep-15	RB1	4889	male	95640	3-43	78
15	150.243	A-12	9-Sep-15	RB1	4407	male	95640	1-41	73
16	150.243	A-18	10-Sep-15	LB2	3808	male	95629	9-39	73
17	150.243	A-19	10-Sep-15	LB2	3719	male	95629	8-48	75
18	150.243	A-20	10-Sep-15	LB2	3619	female	95629	7-37	56
19	150.243	A-21	10-Sep-15	LB2	4515	male	95629	6-26	66
20	150.243	A-22	10-Sep-15	LB2	4517	male	95629	5-45	76.5
21	150.243	A-23	10-Sep-15	LB2	3722	male	95629	4-24	70
22	150.243	A-24	11-Sep-15	LB3		female			68
23	150.243	B-25	12-Sep-15	LB1	4650	male	95629	2-32	66.5
24	150.243	A-26	12-Sep-15	LB1	4520	male	95629	1-31	69
25	150.243	B-28	12-Sep-15	LB1	4997	male	95478	10-50	66
26	150.243	B-29	12-Sep-15	LB1	4639	male	95478	9-39	68
27	150.243	B-30	12-Sep-15	LB1	4491	male	95478	8-48	77



2015 Porcupine River Chum Salmon Telemetry Program - Appendix B – Chum Salmon Radio Tagging Data

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Floy Tag #	Sex	Scale Book Number	Scale Row #	Fork Length (cm)
28	150.243	B-31	12-Sep-15	LB1	4649	male	95477	10-30	67
29	150.283	A-1	12-Sep-15	LB1	4632	male	95477	8-38	64.5
30	150.283	A-2	12-Sep-15	LB1	4289	male	95477	7-27	72
31	150.283	A-3	12-Sep-15	LB1	3150	male			66
32	150.243	A-75	12-Sep-15	LB2	4516	male	95477	9-19	70
33	150.283	A-5	13-Sep-15	LB1	4494	male	95477	6-46	72
34	150.283	A-6	13-Sep-15	LB1	4033	male	95477	5-45	61
35	150.283	A-8	13-Sep-15	LB1	4027	male	95477	4-44	82
36	150.283	A-9	13-Sep-15	LB1	4509	male	95477	3-33	67
37	150.283	A-11	13-Sep-15	LB1	4634	male	95477	2-42	66
38	150.283	A-12	13-Sep-15	LB1	4518	male	95477	1-21	67
39	150.283	A-13	13-Sep-15	LB1		male	95474	10-40	70
40	150.283	A-14	13-Sep-15	LB1	4488	male	95474	9-49	63
41	150.283	A-15	14-Sep-15	LB1	3392	female	95474	8-48	66.5
42	150.283	A-16	14-Sep-15	LB1	4648	male	95474	7-47	66
43	150.283	A-17	14-Sep-15	LB1	3325	male	95474	6-46	68
44	150.283	A-18	14-Sep-15	LB1	4521	male	95474	5-45	71
45	150.283	A-19	14-Sep-15	LB1	4651	female	95474	4-44	68
46	150.283	A-20	14-Sep-15	LB1	4630	male	95474	3-43	81
47	150.283	A-21	14-Sep-15	LB1	4485	female	95474	2-42	71
48	150.283	A-22	14-Sep-15	LB1	4512	male	95474	1-21	71
49	150.283	A-23	14-Sep-15	LB1	4495	male	95465	10-30	72
50	150.283	A-24	14-Sep-15	LB1	4513	male	95465	9-29	73
51	150.283	B-25	14-Sep-15	LB1	4030	female	95465	8-28	65
52	150.283	A-26	15-Sep-15	LB1	4489	male	95465	7-47	68
53	150.283	B-27	15-Sep-15	LB1	4636	male	95465	6-46	68
54	150.283	B-28	15-Sep-15	LB1	4249	male	95465	5-45	67
55	150.283	B-30	15-Sep-15	LB1	4487	female	95465	4-44	73.5
56	150.283	B-31	15-Sep-15	LB1	4996	male	95465	3-43	68.5



2015 Porcupine River Chum Salmon Telemetry Program - Appendix B – Chum Salmon Radio Tagging Data

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Floy Tag #	Sex	Scale Book Number	Scale Row #	Fork Length (cm)
57	150.283	B-32	15-Sep-15	LB1	3711	male	95465	2-42	70
58	150.283	B-33	15-Sep-15	LB1	3708	female	95465	1-41	67
59	150.283	A-75	15-Sep-15	LB1	3340	male	95473	10-50	67
60	150.323	A-01	23-Sep-15	LB2	3486	male	95475	10-50	58
61	150.323	A-02	23-Sep-15	LB2	3638	male	95475	9-49	66
62	150.323	A-04	23-Sep-15	LB2	3642	male	95475	8-48	65
63	150.323	A-03	23-Sep-15	LB2	3628	male	95475	7-47	72
64	150.283	A-4	23-Sep-15	LB2	3482	male	95475	6-46	68
65	150.243	B-32	23-Sep-15	LB2	3637	male	95475	5-45	73
66	150.243	B-33	23-Sep-15	LB2	3484	male	95475	4-44	71
67	150.283	B-29	24-Sep-15	LB2	3709	male	95475	3-43	74
68	150.323	A-14	24-Sep-15	LB2	3629	male	95475	2-42	66
69	150.323	A-08	24-Sep-15	LB2	3636	male	95475	1-41	75
70	150.323	A-15	24-Sep-15	LB2	3480	male	95476	10-50	71
71	150.323	A-09	24-Sep-15	LB2	3632	female	95476	9-49	51
72	150.323	A-13	24-Sep-15	LB2	3902	female	95476	8-48	68
73	150.323	A-11	24-Sep-15	LB2	3641	male	95476	7-47	74
74	150.323	A-12	24-Sep-15	LB2	3710	male	95476	6-46	68
75	150.323	A-06	24-Sep-15	LB2	3634	female	95476	5-45	66
76	150.323	A-05	24-Sep-15	LB2	3626	female	95476	4-44	71
77	150.323	A-16	25-Sep-15	LB2	3639	male	95476	3-33	70
78	150.323	A-19	25-Sep-15	LB2	3640	female	95476	2-22	67
79	150.323	A-17	25-Sep-15	LB2	3630	female	95476	1-11	69
80	150.323	A-18	25-Sep-15	LB2	3637	male	95464	10-50	75
81	150.323	A-20	25-Sep-15	LB2	3633	female	95464	9-49	61
82	150.323	A-23	25-Sep-15	LB2	3483	female	95464	8-48	63
83	150.323	A-21	25-Sep-15	LB2	3485	male	95464	7-47	61
84	150.323	A-26	25-Sep-15	LB2	3635	male	95464	6-46	67
85	150.323	A-22	25-Sep-15	LB2	3374	male	95464	5-45	69



2015 Porcupine River Chum Salmon Telemetry Program - Appendix B – Chum Salmon Radio Tagging Data

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Floy Tag #	Sex	Scale Book Number	Scale Row #	Fork Length (cm)
86	150.323	A-24	25-Sep-15	LB2	3341	male	95464	4-44	65
87	150.323	A-75	26-Sep-15	LB2	3481	male	95463	10-50	72
88	150.323	B-25	26-Sep-15	LB2	3339	female	95463	9-49	62
89	150.323	B-27	26-Sep-15	LB2	3487	male	95463	8-48	66
90	150.323	B-28	26-Sep-15	LB2	3631	male	95463	7-47	67
91	150.323	B-29	26-Sep-15	LB2	3373	male	95463	6-46	63
92	150.323	B-30	26-Sep-15	LB2	3338	female	95463	5-45	66
93	150.323	B-32	26-Sep-15	LB2	3839	male	95472	10-50	63
94	150.323	B-31	26-Sep-15	LB2	3837	male	95472	9-49	72
95	150.323	B-33	26-Sep-15	LB2	3838	male	95472	8-48	69

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**APPENDIX C. LOCATION OF TAGS
DETECTED BY STATIONARY
RADIO TOWERS AND BY
AERIAL TELEMETRY**



Table C1. Summary of radio tagged chum salmon detected by stationary telemetry towers and aerial telemetry flights.

Frequency	Code	Location	Mortality Status	Latitude	Longitude
150.243	A-1	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.52743	-139.25437
150.243	A-2	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.51265	-139.37251
150.243	A-3	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.49785	-139.34228
150.243	A-5	Upper Bell River	Dead	67.69512	-136.64569
150.243	A-6	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52781	-139.32995
150.243	A-8	Migrated downstream after tagging.			
150.243	A-4	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52717	-139.25197
150.243	A-11	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52743	-139.25437
150.243	A-13	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.51265	-139.37251
150.243	A-14	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52494	-139.32743
150.243	A-15	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52844	-139.3174
150.243	A-16	Porcupine River mainstem	Dead	66.75051	-137.93602
150.243	A-17	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.51729	-139.36908
150.243	A-9	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52848	-139.3042
150.243	A-12	Bilwaddy Creek.	Dead	68.17401	-141.11461
150.243	A-18	Porcupine River mainstem	Dead	67.58262	-138.55418
150.243	A-19	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52788	-139.26558
150.243	A-20	Old Crow, and near vicinity	Dead	67.53664	-139.91479
150.243	A-21	Porcupine River mainstem	Dead	66.90394	-137.72012
150.243	A-22	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52851	-139.31119
150.243	A-23	Porcupine River mainstem	Dead	67.53544	-139.22763
150.243	A-24	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52851	-139.31119
150.243	B-25	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50259	-139.36226
150.243	A-26	South Fork of the Fishing Branch River, downstream of the weir	Dead	66.42202	-138.74472
150.243	B-28	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52844	-139.3174
150.243	B-29	South Fork of the Fishing Branch River, upstream of the weir.			
150.243	B-30	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.49928	-139.35
150.243	B-31	Miner River	Dead	66.22321	-138.80196
150.243	A-75	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.50568	-139.36667



2015 Porcupine River Chum Salmon Telemetry Program - Appendix C – Location of Relocated Tags

Frequency	Code	Location	Mortality Status	Latitude	Longitude
150.243	B-32	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.50568	-139.36667
150.243	B-33	Old Crow, and near vicinity	Dead	67.5651	-139.76929
150.283	A-1	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50445	-139.36481
150.283	A-3	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.51807	-139.37009
150.283	A-5	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52822	-139.33387
150.283	A-6	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52847	-139.30266
150.283	A-8	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.50445	-139.36481
150.283	A-9	South Fork of the Fishing Branch River, downstream of the weir	Dead	66.44027	-138.97186
150.283	A-11	South Fork of the Fishing Branch River, downstream of the weir	Dead	66.43463	-138.93991
150.283	A-12	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52799	-139.29958
150.283	A-13	Old Crow, and near vicinity	Dead	67.55494	-139.90516
150.283	A-14	Porcupine River mainstem	Dead	67.16327	-137.65836
150.283	A-15	Old Crow, and near vicinity	Dead	67.55846	-139.89392
150.283	A-16	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.51368	-139.37544
150.283	A-17	Porcupine River mainstem	Dead	67.41449	-137.77091
150.283	A-18	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52847	-139.30266
150.283	A-19	South Fork of the Fishing Branch River, upstream of the weir	Unclear	66.52602	-139.25913
150.283	A-20	South Fork of the Fishing Branch River, downstream of the weir	Dead	66.42836	-138.90424
150.283	A-21	Upper Bell River	Alive	67.71797	-136.76475
150.283	A-22	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.51556	-139.37337
150.283	A-23	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52934	-139.31087
150.283	A-24	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52847	-139.30266
150.283	B-25	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.50277	-139.36218
150.283	A-26	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52934	-139.31087
150.283	B-27	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.51807	-139.37009
150.283	B-28	Porcupine River mainstem	Dead	67.13953	-137.57908
150.283	B-30	South Fork of the Fishing Branch River, downstream of the weir	Unclear	66.42074	-138.85444
150.283	B-31	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50639	-139.36815
150.283	B-32	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50277	-139.36218
150.283	B-33	South Fork of the Fishing Branch River, upstream of the weir.			



2015 Porcupine River Chum Salmon Telemetry Program - Appendix C – Location of Relocated Tags

Frequency	Code	Location	Mortality Status	Latitude	Longitude
150.283	A-75	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.50799	-139.37124
150.283	A-4	South Fork of the Fishing Branch River, downstream of the weir	Dead	66.42671	-138.72189
150.283	B-29		Dead	67.56667	-139.79109
150.323	A-01	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50397	-139.36641
150.323	A-02	Old Crow, and near vicinity	Alive	67.56602	-139.85985
150.323	A-04		Alive	67.56456	-139.8231
150.323	A-03	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.52891	-139.30452
150.323	A-14		Alive	67.56629	-139.85821
150.323	A-15	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52875	-139.31226
150.323	A-09	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.51493	-139.37358
150.323	A-13	South Fork of the Fishing Branch River, downstream of the weir	Unclear	66.48791	-139.08758
150.323	A-11	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52891	-139.30452
150.323	A-12		Dead	67.56525	-139.8155
150.323	A-06	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52799	-139.33224
150.323	A-05	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.52065	-139.36462
150.323	A-16	Old Crow, and near vicinity	Dead	67.57689	-139.63055
150.323	A-19	Old Crow, and near vicinity	Alive	67.56156	-139.73526
150.323	A-17	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50868	-139.3739
150.323	A-18	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.50397	-139.36641
150.323	A-20	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.52881	-139.30967
150.323	A-23	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.53318	-139.25093
150.323	A-26	Upper Bell River	Alive	67.71929	-136.77391
150.323	A-24	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.50868	-139.3739
150.323	B-25	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.52881	-139.30967
150.323	B-27	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.5284	-139.32356
150.323	B-28	Old Crow, and near vicinity	Dead	67.56525	-139.8155
150.323	B-29	Miner River	Alive	66.50177	-138.45562
150.323	B-30	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.52895	-139.295
150.323	B-32	South Fork of the Fishing Branch River, upstream of the weir.	Dead	66.47171	-139.33991
150.323	B-31	South Fork of the Fishing Branch River, upstream of the weir.	Unclear	66.52799	-139.33224



2015 Porcupine River Chum Salmon Telemetry Program - Appendix C – Location of Relocated Tags

Frequency	Code	Location	Mortality Status	Latitude	Longitude
150.323	B-33	South Fork of the Fishing Branch River, upstream of the weir.	Alive	66.50397	-139.36641